

PREFACE FOR FACULTY

Direct form realizations of high-order IIR filters suffer from catastrophic **coefficient quantization sensitivity**: A tiny change in high-order polynomial coefficients causes large displacements of roots, often shifting poles outside the unit circle and causing instability. **Cascade realization** factors the transfer function into a product of Second-Order Sections (SOS / biquads), isolating pole pairs and guaranteeing stability.

Pedagogical Strategy:

1. Demonstrate polynomial root sensitivity: Why 16-bit direct form filters fail for orders $N \geq 4$.
 2. Formulate the Cascade structure: $H(z) = g \prod_{k=1}^K H_k(z)$.
 3. Master **Optimal Pole-Zero Pairing Rules**: Pair complex conjugate poles with the closest transmission zeros in the z -plane to minimize section peak-to-average gain.
 4. Master **Optimal Section Ordering Rules**: Order sections in sequence of increasing Q -factor (increasing pole radius $r \rightarrow 1$) to maximize dynamic range and minimize register overflow.
 5. Formulate L_2 and L_∞ scaling between sections.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Factor** high-order IIR transfer functions into second-order biquad sections (SOS).
 2. **Apply** pole-zero pairing rules to minimize frequency response peaking.
 3. **Order** biquad sections to optimize signal-to-noise ratio (SNR) and prevent overflow.
 4. **Draw** complete Cascade realization signal flow graphs.
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2. MATHEMATICAL FOUNDATIONS

2.1 The Cascade Realization Formulation

$$H(z) = g \prod_{k=1}^K H_k(z) = g \prod_{k=1}^K \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}}, \quad K = \lceil N/2 \rceil$$

Each biquad $H_k(z)$ realizes one real pole pair and one real zero pair:

$$1 + a_{k1}z^{-1} + a_{k2}z^{-2} = (1 - p_k z^{-1})(1 - p_k^* z^{-1}) = 1 - 2r_k \cos(\theta_k)z^{-1} + r_k^2 z^{-2}$$

2.2 Pairing and Ordering Rules

1. **Pole-Zero Pairing:**
 - Pair the complex pole pair p_k having highest Q (closest to unit circle) with the zero pair z_k closest to it in angle θ .
 - This cancels large resonance peaks, keeping the section gain $|H_k(e^{j\omega})|$ relatively flat.
2. **Section Ordering:**
 - Arrange sections in order of **increasing pole radius** r_k (increasing Q / peaking).

- Low- Q sections first $\rightarrow$ High- Q sections last.
- Prevents premature signal saturation in early stages.

3. WORKED NUMERICAL EXAMPLES

Example 18.1: 4th-Order IIR Cascade Realization

Problem: Realize $H(z) = \frac{(1+z^{-1})(1-\sqrt{2}z^{-1}+z^{-2})}{(1-0.9e^{j\pi/4}z^{-1})(1-0.9e^{-j\pi/4}z^{-1})(1-0.6z^{-1})(1+0.5z^{-1})}$ in Cascade Form with optimal pairing.

Solution:

1. Poles:

- Complex pair: $p_{1,2} = 0.9e^{\pm j\pi/4} \Rightarrow 1 - 2(0.9)\cos(\pi/4)z^{-1} + 0.81z^{-2} = 1 - 1.2728z^{-1} + 0.81z^{-2}$. (Radius $r = 0.9$, high Q).
- Real poles: $p_3 = 0.6, p_4 = -0.5 \Rightarrow (1 - 0.6z^{-1})(1 + 0.5z^{-1}) = 1 - 0.1z^{-1} - 0.3z^{-2}$. (Low Q).

2. Zeros:

- Complex pair: $z_{1,2} = 1 \cdot e^{\pm j\pi/4} \Rightarrow 1 - \sqrt{2}z^{-1} + z^{-2} = 1 - 1.4142z^{-1} + z^{-2}$ (at angle $\pi/4$).
- Real zeros: $z_3 = -1 \Rightarrow (1 + z^{-1})$.

3. Optimal Pairing:

- Pair poles at angle $\pi/4$ ($p_{1,2}$) with zeros at angle $\pi/4$ ($z_{1,2}$).
- Section 1 (Low Q): $H_1(z) = \frac{1+z^{-1}}{1-0.1z^{-1}-0.3z^{-2}}$.
- Section 2 (High Q): $H_2(z) = \frac{1-1.4142z^{-1}+z^{-2}}{1-1.2728z^{-1}+0.81z^{-2}}$.

4. Ordering: Section 1 followed by Section 2.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Why are direct form structures not suitable for implementing high-order IIR filters? Explain coefficient quantization sensitivity. (6 Marks) **(b)** Realize the transfer function in Cascade Form:

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1} + \frac{1}{2}z^{-2})}$$

Draw the complete signal flow graph using Direct Form II biquads. (9 Marks)

Model Answer & Step-by-Step Marking Rubric:

• Part (a):

- Root sensitivity explanation: Roots of $A(z)$ depend non-linearly on polynomial coefficients. In high-order polynomials, 1 coefficient perturbation can displace poles by 50 or move them outside $|z| = 1$ (6 Marks)

• Part (b):

- Split into 1st-order and 2nd-order sections: $H_1(z) = \frac{1+\frac{1}{3}z^{-1}}{1-\frac{1}{2}z^{-1}}$, $H_2(z) = \frac{1}{1-\frac{1}{4}z^{-1}+\frac{1}{2}z^{-2}}$ (4 Marks)
- Neatly drawn Cascade SFG of $H_1(z)$ cascaded into $H_2(z)$ using canonical DF-II sections (5 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

b = [1, 1/3]
a = np.convolve([1, -0.5], [1, -0.25, 0.5])
sos = signal.tf2sos(b, a)
print("Computed SOS:")
print(sos)
```